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Machine Learning for Network Intrusion Detection Using the UNSW-NB15 Dataset

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Abstract

The increasing scale and complexity of cyber threats have made intrusion detection an important area of cybersecurity research. Traditional intrusion detection systems often rely on signature-based methods, which can struggle to detect previously unseen or evolving attacks. In response to these limitations, machine learning has emerged as a promising approach for analysing network traffic and identifying malicious behaviour.

This project investigates the effectiveness of non-LLM machine learning models for binary intrusion detection using the UNSW-NB15 dataset. The study focuses on the defensive application of machine learning and evaluates three classification models: Random Forest, Logistic Regression, and Support Vector Machine (SVM). The dataset was preprocessed by removing missing values, encoding categorical variables, and excluding non-informative attributes before training and testing the models. Performance was assessed using accuracy, precision, recall, F1-score, false positive rate, ROC curves, and area under the ROC curve (AUC).

The results showed that all three models were effective at distinguishing malicious from benign traffic. However, Random Forest achieved the strongest overall performance, with an accuracy of 0.958026, precision of 0.956857, recall of 0.982720, F1-score of 0.969616, and ROC-AUC of 0.993702. Logistic Regression and SVM also performed strongly, with SVM achieving the highest recall of 0.996946. Feature importance analysis further showed that the Random Forest model relied on meaningful traffic-related features such as `sttl`, `ct_state_ttl`, `dload`, and `dtttl`.

The findings demonstrate that classical machine learning methods remain highly effective for structured intrusion detection tasks and that Random Forest provided the best overall balance between detection capability and classification reliability on the UNSW-NB15 dataset. The study also highlights the importance of careful preprocessing and methodological validation in producing reliable intrusion detection results.

Chapter 1: Introduction

1.1 Background

Cybersecurity has become an increasingly important concern in modern digital environments due to the growing volume, complexity, and frequency of cyber attacks. Organisations now rely heavily on networked systems to support communication, storage, financial transactions, and critical operations, making them highly vulnerable to malicious activity. As cyber threats continue to evolve, traditional security approaches such as static rule-based detection and signature matching are often no longer sufficient to identify new or sophisticated attacks effectively (**Kaur, Gabrijelcic and Klobucar, 2023; Sommer and Paxson, 2010**).

One of the most important defensive mechanisms in cybersecurity is the intrusion detection system (IDS). An IDS is designed to monitor network or system activity and identify behaviour that may indicate malicious action. Conventional intrusion detection approaches have historically relied on predefined rules or known attack signatures. While these methods can be effective against previously identified threats, they are less capable of detecting unknown, evolving, or non-signature-based attacks (**Sommer and Paxson, 2010**).

In response to these limitations, machine learning has become an increasingly significant area of research in intrusion detection. Machine learning models can analyse large volumes of network traffic data and learn patterns that distinguish malicious activity from benign behaviour. This allows them to detect suspicious traffic based on behavioural characteristics rather than relying only on manually defined signatures. As a result, machine learning offers the potential to improve detection accuracy, increase adaptability, and support more intelligent cybersecurity defence (**Kaur, Gabrijelcic and Klobucar, 2023**).

1.2 Problem Context

The growing use of artificial intelligence in cybersecurity has created both opportunities and challenges. On the defensive side, machine learning techniques are now widely explored for tasks such as anomaly detection, threat classification, malware analysis, and automated response support. These approaches are attractive because they can process large-scale data more efficiently than manual analysis and can identify complex relationships in data that may not be obvious to human operators (**Kaur, Gabrijelcic and Klobucar, 2023**).

However, there are still important questions about how effective these models are in realistic intrusion detection settings. Different machine learning models may perform differently depending on the structure of the dataset, the type of features used, and the balance between false positives and false negatives. In cybersecurity, this trade-off is especially important. A model that misses attacks may leave systems exposed, while a model that generates too many false alarms may overwhelm analysts and reduce trust in the detection system (**Sommer and Paxson, 2010**).

This creates a practical need to evaluate and compare machine learning approaches in a structured way. Rather than assuming that all AI-based detection systems are equally effective, it is necessary to

identify which techniques provide the best balance between detection capability, reliability, and interpretability.

1.3 Rationale for the Study

This project focuses on the defensive use of non-LLM machine learning techniques in cybersecurity, specifically in the context of network intrusion detection. The study was designed to examine whether classical machine learning models can effectively classify malicious and benign network traffic and to compare their relative performance using a benchmark intrusion detection dataset.

The rationale for this project is based on three main considerations. First, intrusion detection remains a core cybersecurity challenge, particularly as attacks become more varied and less dependent on known signatures. Second, classical machine learning methods remain highly relevant for structured cybersecurity datasets and are often more practical and interpretable than more computationally intensive approaches. Third, comparative evaluation is necessary because different models offer different strengths, and the most suitable model depends on operational priorities such as attack detection sensitivity, false positive reduction, and interpretability (**Kaur, Gabrijelcic and Klobucar, 2023**).

By comparing multiple machine learning models on the same dataset and evaluation framework, this project provides a clearer understanding of how effective these techniques are for intrusion detection and which model offers the most appropriate balance of performance.

1.4 Aim

The aim of this project is to investigate the use of non-LLM machine learning techniques for network intrusion detection using the UNSW-NB15 dataset, and to evaluate the effectiveness of selected models in distinguishing between normal and malicious network traffic.

1.5 Objectives

To achieve this aim, the project was guided by the following objectives:

1. To explore and preprocess the UNSW-NB15 dataset for binary intrusion detection.
2. To implement machine learning models for classifying benign and malicious traffic.
3. To evaluate the performance of the selected models using standard classification metrics.
4. To compare the models in order to identify the most effective approach for the dataset used.
5. To analyse the practical implications and limitations of machine learning-based intrusion detection.

1.6 Research Question

The main research question addressed in this study is:

How effectively can non-LLM machine learning models detect malicious network traffic in a binary intrusion detection setting?

A secondary question explored through model comparison is:

Which of the selected machine learning models provides the best balance between attack detection performance and classification reliability on the UNSW-NB15 dataset?

1.7 Scope of the Project

This project is limited to the defensive application of machine learning for binary intrusion detection. The study focuses specifically on detecting whether network traffic is normal or malicious, rather than identifying the exact attack category. Although the UNSW-NB15 dataset includes multiple attack classes, the classification task in this project was intentionally restricted to binary detection in order to maintain a clear and achievable scope (Moustafa and Slay, 2015).

The project also focuses only on classical non-LLM machine learning techniques. The models implemented were Random Forest, Logistic Regression, and Support Vector Machine. More advanced approaches such as deep learning, hybrid detection systems, or real-time deployment architectures were outside the scope of this study.

The evaluation was conducted using an offline benchmark dataset rather than live network traffic. As a result, the findings reflect controlled experimental performance rather than direct operational deployment (Sommer and Paxson, 2010).

1.8 Overview of Method

To investigate the research question, the project followed a structured experimental workflow. First, the UNSW-NB15 dataset was explored to understand its structure, feature distribution, and class balance. The data was then preprocessed by removing missing values, encoding categorical variables, and preparing the feature set for model training.

Three machine learning models were selected for evaluation: Random Forest, Logistic Regression, and Support Vector Machine. These models were chosen to represent different types of machine learning approaches, including ensemble tree-based learning, linear modelling, and margin-based classification. After training and testing the models, their performance was evaluated using metrics such as accuracy, precision, recall, F1-score, false positive rate, ROC curves, and area under the ROC curve (AUC).

The final stage of the study involved comparing the models and interpreting the results in relation to practical intrusion detection requirements.

1.9 Significance of the Study

This project is significant because it contributes to the understanding of how classical machine learning techniques can support intrusion detection in cybersecurity. While artificial intelligence is often discussed in broad terms, practical evaluation is needed to determine how specific models behave in realistic security tasks.

The study is also valuable because it shows that model performance must be interpreted carefully rather than assumed. Different models may excel in different areas; for example, one model may achieve the highest recall while another provides better overall balance across evaluation metrics. This has direct relevance for real-world cybersecurity practice, where the choice of model may depend on whether the priority is maximum attack capture, reduced false alarms, or better interpretability (Sommer and Paxson, 2010).

In addition, the project demonstrates the importance of methodological care in machine learning experiments, particularly in relation to preprocessing, feature selection, and evaluation design. This strengthens the academic value of the work and supports more reliable interpretation of machine learning performance in intrusion detection.

1.10 Dissertation Structure

The remainder of this dissertation is organised as follows:

- **Chapter 2** reviews the existing literature on intrusion detection, machine learning in cybersecurity, and related benchmark datasets.
- **Chapter 3** describes the methodology used in the project, including dataset selection, preprocessing, model implementation, and evaluation metrics.
- **Chapter 4** presents the experimental results for the Random Forest, Logistic Regression, and Support Vector Machine models.
- **Chapter 5** discusses the findings, interprets their significance, and considers the practical implications and limitations of the study.
- **Chapter 6** concludes the dissertation by summarising the key findings and suggesting directions for future work.

1.11 Chapter Summary

This chapter introduced the background and motivation for the project, outlined the research problem, and explained the relevance of machine learning for intrusion detection. It also defined the aim, objectives, research questions, scope, and significance of the study, and provided an overview of the dissertation structure. The next chapter reviews the literature relevant to intrusion detection systems, machine learning approaches in cybersecurity, and the benchmark datasets used in this field.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews the literature relevant to machine learning-based intrusion detection and situates the project within the wider field of cybersecurity research. It begins by discussing intrusion detection as a core defensive cybersecurity function and the limitations of traditional signature-based approaches. It then examines how machine learning has been applied to intrusion detection, with particular attention to classical non-LLM techniques such as Random Forest, Logistic Regression, and Support Vector Machines. The chapter also considers the importance of benchmark datasets, particularly UNSW-NB15, and reviews key concerns raised in the literature regarding realism, false positives, and the gap between experimental performance and real-world deployment.

The purpose of this review is to establish the academic context of the project and to justify the selection of dataset, models, and evaluation strategy used in the study.

2.2 Intrusion Detection in Cybersecurity

Intrusion detection systems (IDS) are an important component of modern cybersecurity defence. Their main role is to monitor network or system activity and identify behaviour that may indicate malicious actions such as unauthorised access, exploitation, denial of service, or abnormal communication patterns. IDS technologies are generally used to strengthen visibility within digital environments and to provide early warning of possible attacks.

Traditional IDS approaches have relied heavily on signature-based or rule-based detection. These systems compare observed traffic or behaviour against predefined attack signatures or manually designed detection rules. Signature-based approaches can be highly effective for detecting known threats, especially where attack patterns are already documented. However, they are less effective when facing new, evolving, or obfuscated attacks that do not match known signatures. This weakness has become increasingly significant as cyber threats have grown in volume, sophistication, and diversity (**Sommer and Paxson, 2010**).

As a result, research has increasingly shifted toward behaviour-based and anomaly-based detection methods. These approaches attempt to identify suspicious activity by examining patterns in data rather than relying solely on known attack definitions. Machine learning has become particularly attractive in this context because it allows systems to learn patterns from data and make predictions about whether traffic is benign or malicious (**Sommer and Paxson, 2010; Kaur, Gabrijelcic, and Klobucar, 2023**).

2.3 Artificial Intelligence and Machine Learning in Cybersecurity

Artificial intelligence has become increasingly important in cybersecurity due to its ability to process large-scale data, identify hidden patterns, and support faster analysis than purely manual approaches. In practical terms, machine learning has been applied to areas such as intrusion detection, malware

classification, phishing detection, threat intelligence, and automated alert triage (Kaur, Gabrijelčič, and Klobučar, 2023).

A major strength of machine learning in cybersecurity is its ability to detect patterns that are difficult to define explicitly using handcrafted rules. For intrusion detection, this is particularly useful because network traffic often contains subtle behavioural differences between malicious and benign activity. Machine learning models can learn these distinctions from training data and then apply them to unseen traffic (Kaur, Gabrijelčič, and Klobučar, 2023).

Kaur, Gabrijelčič, and Klobučar describe artificial intelligence as a growing and significant area in cybersecurity research, with applications across a wide range of defensive tasks. Their review highlights the promise of AI-based methods while also identifying limitations and future research needs. This broader perspective is important because it shows that AI in cybersecurity should not be viewed as a single solution, but rather as a family of techniques with different strengths, weaknesses, and application contexts (Kaur, Gabrijelčič, and Klobučar, 2023).

For this reason, the present project does not attempt to treat “AI” as a single concept. Instead, it focuses specifically on classical non-LLM machine learning techniques in intrusion detection and examines how different models perform when applied to structured network traffic data.

2.4 Machine Learning for Intrusion Detection

Machine learning-based intrusion detection has become one of the most active research areas in cybersecurity. The general goal is to train models to classify traffic as benign or malicious based on observed features such as packet sizes, protocol behaviour, state information, and flow-level statistics (Kaur, Gabrijelčič, and Klobučar, 2023).

These approaches are typically divided into supervised and unsupervised learning. Supervised learning uses labelled data, where examples of both benign and malicious traffic are provided during training. Unsupervised learning attempts to identify anomalies without fully labelled attack classes. The present project uses supervised learning, as the UNSW-NB15 dataset provides labelled examples of attack and normal traffic (**Moustafa and Slay, 2015**).

The literature shows that machine learning-based IDS can achieve high performance on benchmark datasets. However, performance depends heavily on the choice of model, feature representation, preprocessing strategy, and dataset quality. As a result, many studies compare multiple algorithms in order to determine which model is most appropriate for a specific dataset or intrusion detection setting (**Sommer and Paxson, 2010**).

A recurring theme in this literature is that no single algorithm is universally best. Instead, models must be judged in relation to the structure of the data and the priorities of the detection task. In cybersecurity, this often means balancing high attack detection rates against the practical need to keep false alarms manageable (Sommer and Paxson, 2010).

2.5 Random Forest in Intrusion Detection

Random Forest is one of the most widely used classical machine learning models in intrusion detection research. It is an ensemble learning method that combines the outputs of multiple decision trees to improve predictive performance and reduce overfitting. Because it can capture complex, non-linear relationships between variables, it is especially well suited to structured tabular cybersecurity datasets (Kaur, Gabrijelčič, and Klobučar, 2023).

A major advantage of Random Forest is that it can handle datasets containing many features with different scales and types. This is useful in intrusion detection because network datasets often include a mixture of statistical traffic measures, protocol attributes, and derived behavioural features. Random Forest is also relatively robust to noise and can provide feature importance measures, which improve interpretability by showing which variables contributed most strongly to classification decisions (Kaur, Gabrijelčič, and Klobučar, 2023).

In many intrusion detection studies, Random Forest performs strongly because attack behaviour is often defined not by a single feature but by combinations of characteristics. Ensemble tree-based methods are able to model these interactions more effectively than purely linear classifiers. This helps explain why Random Forest is frequently reported as one of the strongest classical models for intrusion detection tasks (Kaur, Gabrijelčič, and Klobučar, 2023).

The present study includes Random Forest as its primary model partly for these reasons. It offers strong predictive performance, robustness, and a level of interpretability that is particularly useful in cybersecurity research.

2.6 Logistic Regression as Baseline Model

Logistic Regression is a common baseline model in classification research and is frequently used in intrusion detection studies for comparison purposes. It is a linear model that estimates the probability that an input belongs to a particular class. Because it is simpler than many other machine learning algorithms, it is often used to establish a benchmark against which more advanced models can be evaluated.

The value of Logistic Regression in IDS research lies in its interpretability and simplicity. It is computationally efficient, relatively easy to implement, and provides a useful reference point for judging whether more complex models actually provide meaningful improvement. If a simple linear classifier performs nearly as well as an ensemble method, then the additional complexity of the larger model may be harder to justify (Sommer and Paxson, 2010).

However, Logistic Regression has limitations when applied to intrusion detection. Network traffic data often contains non-linear interactions and complex feature dependencies, which linear models may struggle to capture fully. For this reason, Logistic Regression is often expected to perform adequately, but not as strongly as models that can represent more complex decision boundaries (Sommer and Paxson, 2010).

Its inclusion in this project is therefore important from a methodological perspective: it provides a baseline against which the value of stronger non-linear models can be assessed.

2.7 Support Vector Machine in Intrusion Detection

Support Vector Machines (SVMs) are also commonly used in intrusion detection research. SVMs work by identifying a decision boundary that maximises the margin between classes, making them powerful classifiers in many high-dimensional settings. They are particularly attractive because they can perform well even when the relationship between features is not simple.

In intrusion detection, SVMs are often valued for their ability to produce strong classification performance, particularly in terms of recall. This can be important in cybersecurity contexts because high recall means that fewer attacks are missed. However, SVM performance depends strongly on parameter choice, kernel type, and dataset scale. On large datasets, standard SVM implementations can become computationally expensive, which is why practical studies often use linear approximations such as LinearSVC for efficiency (**Sommer and Paxson, 2010**).

The literature generally presents SVM as a strong alternative to ensemble methods, though often with trade-offs. In some cases, SVMs may be highly sensitive in attack detection, but less balanced overall in terms of precision or operational efficiency. This makes them useful for comparison in projects such as this one, where model trade-offs are central to the analysis (**Sommer and Paxson, 2010**).

2.8 The Importance of Evaluation in IDS Research

Evaluation metrics are especially important in intrusion detection research because simple accuracy is often not sufficient to judge model quality. In cybersecurity, the cost of different errors is not equal. Missing an attack can be highly dangerous, while incorrectly flagging benign activity can create unnecessary workload for analysts (**Sommer and Paxson, 2010**).

For this reason, studies commonly report precision, recall, F1-score, confusion matrices, false positive rate, and ROC-AUC. Each metric captures a different aspect of model behaviour. Recall is important because it measures how many attacks are successfully detected. Precision is important because it reflects how often flagged attacks are actually malicious. The F1-score provides a balance between these two. False positive rate is operationally important because too many false alarms can reduce the usefulness of the system. ROC curves and AUC provide a broader view of discrimination performance across thresholds (**Sommer and Paxson, 2010**).

This literature supports the evaluation design used in the present project. Rather than relying only on accuracy, the project uses a set of complementary metrics in order to assess the strengths and trade-offs of each model more meaningfully.

2.9 Benchmark Datasets for Intrusion Detection

The quality of machine learning-based intrusion detection research depends heavily on the dataset used. Datasets provide the basis on which models are trained, tested, and compared, so their realism and structure have a major influence on reported performance.

Historically, datasets such as KDD99 and NSL-KDD were widely used in intrusion detection research. However, these datasets have been criticised for being outdated and not fully representative of modern network behaviour. This has led researchers to favour more recent benchmark datasets that include more realistic traffic and more diverse attack patterns (**Sharafaldin, Lashkari and Ghorbani, 2018; Moustafa and Slay, 2015**).

Sharafaldin, Lashkari, and Ghorbani proposed CICIDS2017 as a more realistic modern intrusion detection dataset, while Moustafa and Slay introduced UNSW-NB15 as a comprehensive benchmark for intrusion detection systems. Both datasets are regarded as important improvements over older benchmarks, though they differ in structure, feature sets, and attack scenarios (**Sharafaldin, Lashkari and Ghorbani, 2018; Moustafa and Slay, 2015**).

The present project uses UNSW-NB15 because it offers a useful balance between realism, feature richness, and practical usability. It contains a diverse set of modern traffic features and multiple attack types while remaining manageable for classical machine learning experimentation (**Moustafa and Slay, 2015**).

2.10 Significance of UNSW-NB15

UNSW-NB15 has become a widely used benchmark dataset in intrusion detection research. It was designed to overcome some of the limitations of older datasets by including more modern attack scenarios and more realistic synthetic traffic generation. The dataset includes both normal traffic and multiple categories of malicious activity, as well as a large set of network and flow-related features (**Moustafa and Slay, 2015**).

One of the main reasons UNSW-NB15 is significant is that it includes a combination of packet-level and behaviour-oriented features. This makes it suitable for machine learning because models can learn not only simple signatures but also more complex traffic patterns. It is also large enough to support robust experimentation while still being manageable in a university project setting (**Moustafa and Slay, 2015**).

The use of UNSW-NB15 in this dissertation is therefore justified both practically and academically. It aligns the study with established IDS research practice while also providing a realistic enough setting to support meaningful model comparison.

2.11 Realism and the Closed-World Problem

Although machine learning has produced strong results in intrusion detection research, a major concern in the literature is that benchmark performance does not always translate into operational success. Sommer and Paxson's well-known critique of machine learning for intrusion detection argues

that many IDS experiments assume a “closed world” in which training and testing conditions are artificially well-defined. In real-world environments, attackers adapt, network behaviour changes, and class boundaries may not remain stable (**Sommer and Paxson, 2010**).

This critique is highly important for the present study. It means that strong performance on UNSW-NB15 should not automatically be interpreted as evidence that the model would perform equally well in production. Rather, the results should be understood as evidence of model capability in a controlled benchmark setting (**Sommer and Paxson, 2010**).

This does not reduce the value of the project, but it does shape how the findings should be interpreted. The goal is not to claim full deployment readiness, but to evaluate model performance in a structured and academically recognised intrusion detection scenario.

2.12 AI as a Defensive and Offensive Force

The broader literature also recognises that AI is not only a defensive tool. Brundage et al. discuss the malicious use of artificial intelligence and show that AI can also support offensive cyber activity. More recently, Marchal et al. propose a taxonomy of generative AI misuse, identifying the ways in which AI can support reconnaissance, automation, and attack decision-making (**Brundage et al., 2018; Marchal et al., 2024**).

Although the present project focuses on defensive intrusion detection rather than live offensive implementation, this literature remains relevant because it frames the wider significance of AI in cybersecurity. If attackers also benefit from AI-assisted methods, then improving defensive detection becomes even more important. It also means that defensive models should not be assessed in isolation from the broader evolution of attacker behaviour (**Brundage et al., 2018; Marchal et al., 2024**).

This supports the wider rationale of the dissertation by showing that machine learning-based intrusion detection is part of a larger cybersecurity landscape in which both defence and offence are increasingly influenced by AI.

2.13 Research Gap and Justification for the Present Study

The literature shows that machine learning has clear potential in intrusion detection and that benchmark datasets such as UNSW-NB15 are widely used for evaluating performance. It also shows that Random Forest, Logistic Regression, and SVM are all relevant models within IDS research. However, the literature also highlights important issues such as model trade-offs, false positive costs, and the gap between benchmark experiments and real-world deployment (**Kaur, Gabrijelcic, and Klobucar, 2023; Sommer and Paxson, 2010; Moustafa and Slay, 2015**).

This creates a clear justification for the present study. There is value in conducting a structured comparison of classical machine learning models on a modern benchmark dataset, especially when the evaluation goes beyond simple accuracy and considers practical metrics such as recall, false positive rate, ROC-AUC, and interpretability through feature importance.

The study therefore contributes by providing a focused evaluation of non-LLM machine learning for binary intrusion detection, using a clearly defined dataset, a transparent preprocessing pipeline, and a comparative model framework. In doing so, it helps clarify which model provides the strongest balance of performance for the specific task investigated.

2.14 Chapter Summary

This chapter has reviewed the literature on intrusion detection, machine learning in cybersecurity, model selection, evaluation metrics, and benchmark datasets. The review has shown that machine learning offers significant potential for intrusion detection, but that results depend heavily on model choice, preprocessing strategy, dataset quality, and evaluation design. It has also shown that benchmark success must be interpreted carefully in light of concerns about realism and deployment.

The literature supports the use of Random Forest, Logistic Regression, and SVM as suitable comparative models and justifies the selection of UNSW-NB15 as an appropriate dataset for this project. The next chapter explains the methodology used to implement and evaluate these models in the present study.

Chapter 3: Methodology

3.1 Research Approach

This project adopted an experimental machine learning approach to investigate the effectiveness of non-LLM artificial intelligence techniques for network intrusion detection. The study focused on the defensive use of machine learning by training and evaluating multiple classification models on a benchmark intrusion detection dataset. The objective was to determine how well these models could distinguish between normal and malicious network traffic and to compare their performance using standard evaluation metrics.

The methodology was structured into four main stages: data exploration, data preprocessing, model training, and comparative evaluation. This workflow was implemented across a set of Jupyter notebooks in order to maintain a clear and reproducible experimental pipeline.

3.2 Dataset Selection

The dataset used in this study was the **UNSW-NB15** intrusion detection dataset. This dataset was selected because it is widely used in cybersecurity research and provides a more modern representation of network traffic and attack behaviour than older benchmark datasets such as KDD99 and NSL-KDD. It contains both benign and malicious traffic records, making it suitable for supervised machine learning classification tasks (**Moustafa and Slay, 2015**).

For this project, the **training-set CSV file** from UNSW-NB15 was used. The dataset contained **175,341 records and 45 columns**, including both numerical and categorical network traffic features (Moustafa and Slay, 2015). It also included the fields `label`, and `attack_cat`, where `label` represents binary classification between normal and attack traffic, and `attack_cat` provides multi-class attack category information.

The dataset was appropriate for this study because it includes a broad range of network attributes such as protocol type, service, state, packet-level features, and traffic flow statistics. These variables allow machine learning models to learn patterns associated with malicious activity in realistic network traffic scenarios.

3.3 Exploratory Data Analysis

An exploratory data analysis phase was conducted before preprocessing and model training. This stage was used to understand the structure and characteristics of the dataset and to identify any issues that could affect model performance.

The dataset was first loaded into a pandas DataFrame and inspected using summary functions such as `head()`, `info()`, and `describe()`. Missing values were checked using column-wise null counts. In addition, several visualisations were generated to better understand the dataset:

- a count plot showing the distribution of normal versus attack traffic using the `label` field
- a count plot showing the distribution of attack categories using the `attack_cat` field
- a protocol distribution plot using the `proto` feature
- a correlation heatmap for numerical variables

This stage helped confirm that the dataset was suitable for binary intrusion detection and provided insight into class distribution and feature relationships prior to preprocessing.

3.4 Data Preprocessing

Data preprocessing was carried out to transform the raw dataset into a format suitable for machine learning models. The main preprocessing steps were as follows.

3.4.1 Handling Missing Values

Missing values were removed using `dropna()`. This ensured that incomplete records did not interfere with model training or evaluation. After this step, the dataset shape was checked again to confirm the cleaned data size.

3.4.2 Encoding Categorical Variables

Because machine learning algorithms require numerical input, all categorical variables were converted into numeric form. Categorical columns were identified by selecting features with the `object` data type. A `LabelEncoder` was then applied to each categorical column individually.

This step transformed textual values such as protocol names, service names, and connection states into integer-encoded representations that could be processed by the models.

3.4.3 Feature and Target Definition

The dataset was prepared for binary classification by separating input features from the target variable. The `label` column was used as the target variable, where `0` represented normal traffic and `1` represented attack traffic.

The `attack_cat` column was excluded from the input features because it provides attack category information that was not required for binary classification and could introduce redundancy. In addition, the `id` column was removed because it functions only as a record identifier and does not represent a meaningful network traffic feature. Therefore, the feature matrix `X` was created by dropping `id`, `label`, and `attack_cat`, while the target vector `y` was defined using `label`.

3.4.4 Train-Test split

The processed dataset was divided into training and testing subsets using `train_test_split()` from scikit-learn. A standard **80:20 split** was used, with `random_state=42` to ensure reproducibility.

The training data was used to fit the models, while the testing data was reserved for performance evaluation on unseen examples. This approach helps assess how well the models generalise beyond the data they were trained on.

3.4.5 Feature Scaling

Feature scaling was applied in the model comparison stage for models that are sensitive to input magnitude, specifically Logistic Regression and Support Vector Machines. A `StandardScaler` was used to standardise the training data and then transform the test data using the same fitted scaler.

Scaling was not applied to the Random Forest model, as tree-based algorithms are generally insensitive to feature scale.

3.5 Model Selection

Three machine learning models were used in this study:

3.5.1 Random Forest

Random Forest was used as the primary model for intrusion detection. It is an ensemble learning algorithm that constructs multiple decision trees and combines their outputs to improve classification performance. Random Forest was selected because it is well suited to tabular cybersecurity datasets,

can model non-linear feature relationships, and provides feature importance scores that support interpretability (Kaur, Gabrijelčič, and Klobučar, 2023).

A tuned Random Forest configuration was used with the following parameters:

- `n_estimators = 300`
- `max_depth = 20`
- `min_samples_split = 5`
- `min_samples_leaf = 2`
- `n_jobs = -1`
- `random_state = 42`

These settings were used to improve predictive performance while reducing overfitting and improving computational efficiency.

3.5.2 Logistic Regression

Logistic Regression was included as a baseline linear model. It provides a useful reference point because it is simple, widely understood, and commonly used in classification tasks. Although it may not capture complex non-linear patterns as effectively as ensemble methods, it offers a strong benchmark for comparison.

The model was trained with `max_iter = 1000` to ensure convergence.

3.5.3 Support Vector Machines as a Comparative Model

A linear Support Vector Machine was also included in the comparison. Initially, a standard SVM approach was considered, but due to computational cost on a dataset of this size, `LinearSVC` was selected instead. This made training feasible while still allowing the project to compare a margin-based classifier against the other models.

The use of `LinearSVC` also made the evaluation more practical for a final-year project environment while preserving the comparative value of the model.

3.6 Model Training

The Random Forest model was trained in a dedicated notebook after the processed dataset was loaded and the data was split into training and testing sets. Predictions were then generated on the test set for evaluation.

The Logistic Regression and Linear SVM models were trained in a separate comparison notebook using the same train-test split. For these models, scaled training and test sets were used. The Random Forest model was then loaded into the comparison notebook so that all three models could be evaluated on the same test data.

This ensured consistency in the comparison and allowed all performance metrics to be calculated under the same experimental conditions.

3.7 Evaluation Metrics

Model performance was assessed using several classification metrics commonly used in intrusion detection research (**Sommer and Paxson, 2010**).

3.7.1 Classification Report

For each model, a classification report was generated to obtain:

- precision
- recall
- F1-score
- support

These metrics provide a balanced view of classification performance, especially in cybersecurity contexts where the cost of missed attacks and false alarms can differ significantly (**Sommer and Paxson, 2010**).

3.7.2 Confusion Matrix

A confusion matrix was used to visualise model predictions in terms of:

- true positives
- true negatives
- false positives
- false negatives

This was particularly useful for understanding how well the Random Forest model distinguished normal traffic from attacks.

3.7.3 False Positive Rate

The False Positive Rate (FPR) was calculated for the Random Forest model using the confusion matrix. This metric is important in intrusion detection because a high false positive rate can lead to alert fatigue and reduce the practical usefulness of a detection system (**Sommer and Paxson, 2010**).

3.7.4 ROC Curve and AUC

Receiver Operating Characteristic (ROC) curves were generated to compare the discrimination ability of the models. For Random Forest and Logistic Regression, probability outputs were used. For the Linear SVM model, the `decision_function()` output was used because `LinearSVC` does not provide class probabilities.

Area Under the Curve (AUC) was calculated for each model to summarise its ability to distinguish between benign and malicious traffic. The ROC comparison plot provided a visual basis for identifying the strongest overall model.

3.7.5 Feature Importance

For the Random Forest model, feature importance values were extracted and ranked. The most important features were visualised in a horizontal bar chart. This step added interpretability to the analysis by identifying which network traffic attributes contributed most strongly to intrusion detection.

3.8 Experimental Workflow

The full experimental workflow used in this project can be summarised as follows:

1. Load and inspect the UNSW-NB15 dataset
2. Explore class distribution, protocol distribution, and feature correlations
3. Remove missing values
4. Encode categorical variables using label encoding
5. Define feature matrix and target label
6. Split data into training and test sets
7. Train the Random Forest model
8. Evaluate Random Forest using classification metrics, confusion matrix, false positive rate, ROC curve, and feature importance
9. Scale the data for linear models
10. Train Logistic Regression and Linear SVM
11. Compare all three models using ROC curves, AUC scores, and a performance table

This staged workflow ensured that the project remained organised, reproducible, and aligned with the research objective of evaluating machine learning approaches for intrusion detection.

3.9 Reproducibility and Implementation Environment

The implementation was carried out in Python using Jupyter notebooks. The main libraries used in the project included:

- pandas
- numpy
- matplotlib
- seaborn
- scikit-learn
- joblib

To improve reproducibility, a fixed random seed (`random_state=42`) was used in train-test splitting and in the Random Forest model. In addition, the trained Random Forest model and results

tables were saved to disk so that the evaluation could be reused in later stages of the project without retraining the model.

3.10 Methodology Limitations

Although the methodology provides a practical and structured evaluation of machine learning for intrusion detection, several limitations should be acknowledged. First, the study uses a benchmark dataset rather than live network traffic, meaning the results reflect offline experimental conditions rather than real-time deployment (**Sommer and Paxson, 2010**). Second, label encoding was used for categorical variables, which is computationally simple but may not capture all relationships between categories. Third, the comparison was limited to three classical machine learning models, so the study does not evaluate deep learning or streaming detection approaches.

Despite these limitations, the methodology is appropriate for the project scope and provides a reliable basis for evaluating non-LLM machine learning models in intrusion detection.

Chapter 4: Results

4.1 Introduction

This chapter presents the results of the machine learning experiments carried out on the UNSW-NB15 dataset for binary intrusion detection. The purpose of the evaluation was to determine how effectively the selected machine learning models could distinguish between normal and malicious network traffic. The chapter begins with the performance of the primary Random Forest model, followed by analysis of the confusion matrix, ROC curve, and feature importance results. A comparative evaluation is then presented for Random Forest, Logistic Regression, and Support Vector Machine (SVM), allowing the relative strengths of the models to be assessed.

4.2 Dataset Overview

The experiments were conducted using the processed UNSW-NB15 training dataset, which contained 175,341 records and 45 columns before feature-target separation. After preprocessing, the dataset was used for binary classification, where the `label` field represented normal traffic and malicious traffic. The `attack_cat` field was excluded from the feature set because the task focused on binary intrusion detection rather than multi-class attack classification.

The dataset provided a suitable basis for experimental evaluation because it contained a mixture of numerical and encoded categorical network traffic features. These features represented protocol behaviour, service information, state transitions, packet statistics, and flow-level attributes, all of which are relevant to intrusion detection. The class distribution showed that attack traffic was more frequent than normal traffic, indicating a degree of class imbalance that is important when interpreting model performance.

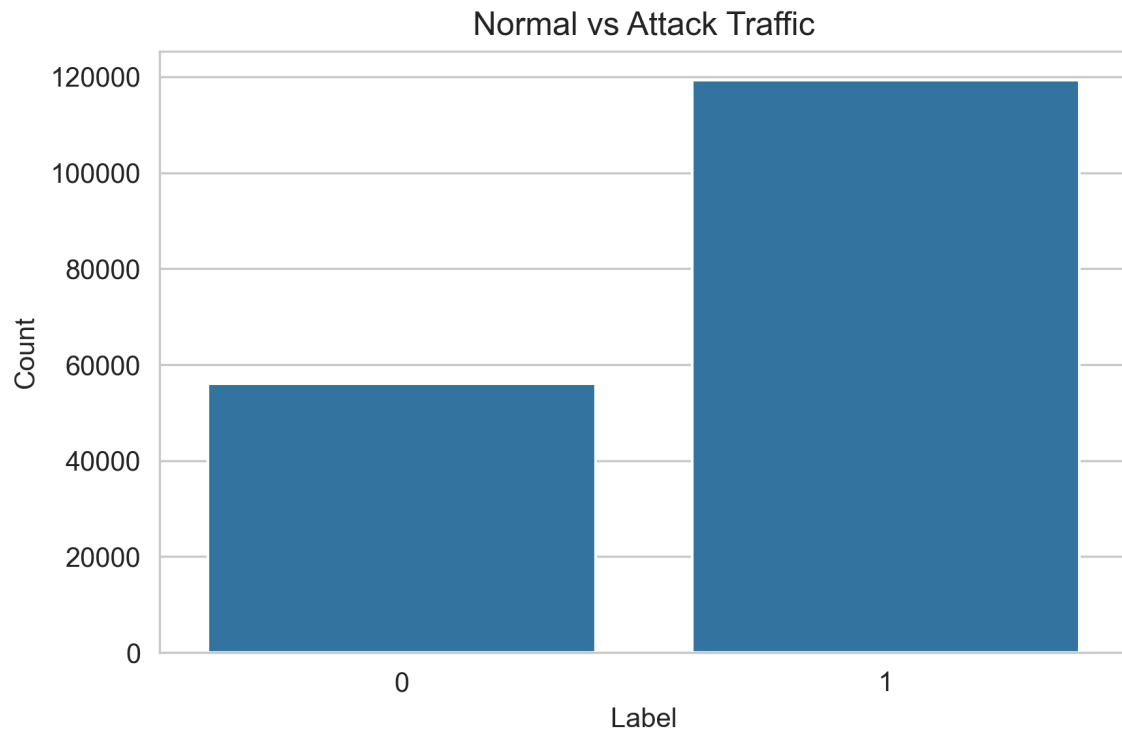


Figure 4.1: Distribution of benign and malicious traffic

4.3 Random Forest Model Performance

The first model evaluated was the Random Forest classifier. This model served as the primary intrusion detection model in the study due to its strong suitability for tabular cybersecurity datasets and its ability to model non-linear relationships between features.

After training, the Random Forest model was evaluated on the test set using standard classification metrics. The model achieved an **accuracy of 0.958026**, **precision of 0.956857**, **recall of 0.982720**, and **F1-score of 0.969616**. These results indicate strong overall performance in distinguishing malicious traffic from benign traffic.

The high recall is particularly important in intrusion detection, as it indicates that the model successfully detected most malicious instances in the dataset. At the same time, the high precision shows that the majority of traffic flagged as malicious was correctly identified. Together, these values suggest that the Random Forest model provided a strong balance between attack detection capability and classification reliability.

Overall, the Random Forest classifier produced the strongest all-round performance of the evaluated models and was therefore retained as the main detection model in this study.

4.4 Confusion Matrix Analysis

The confusion matrix for the Random Forest model provided a detailed breakdown of prediction outcomes in terms of true positives, true negatives, false positives, and false negatives.

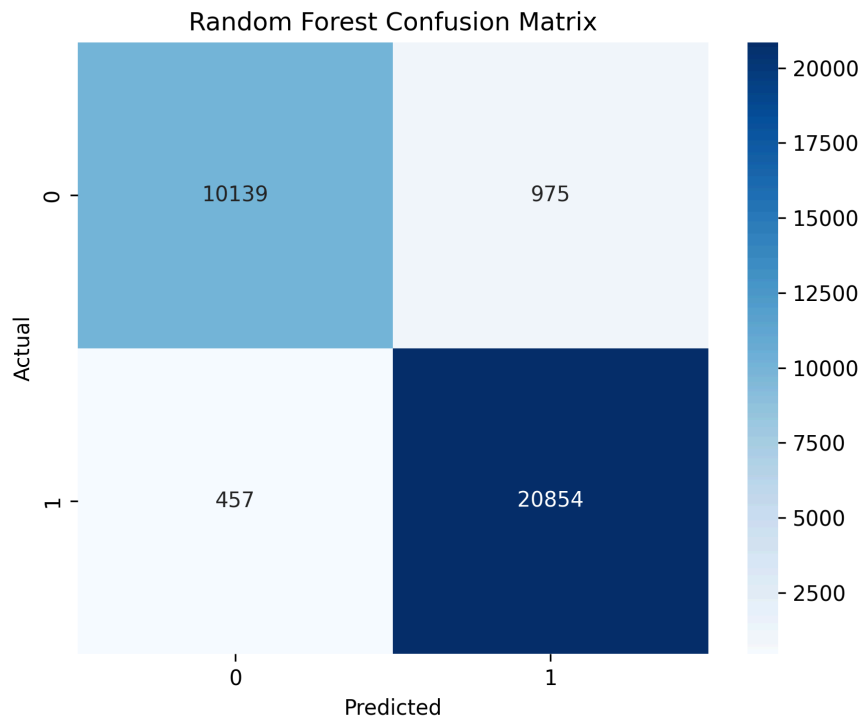


Figure 4.2: Confusion Matrix for the Random Forest intrusion detection model

The confusion matrix showed:

- **True Negatives (TN)** = 10,139
- **False Positives (FP)** = 975
- **False Negatives (FN)** = 457
- **True Positives (TP)** = 20,854

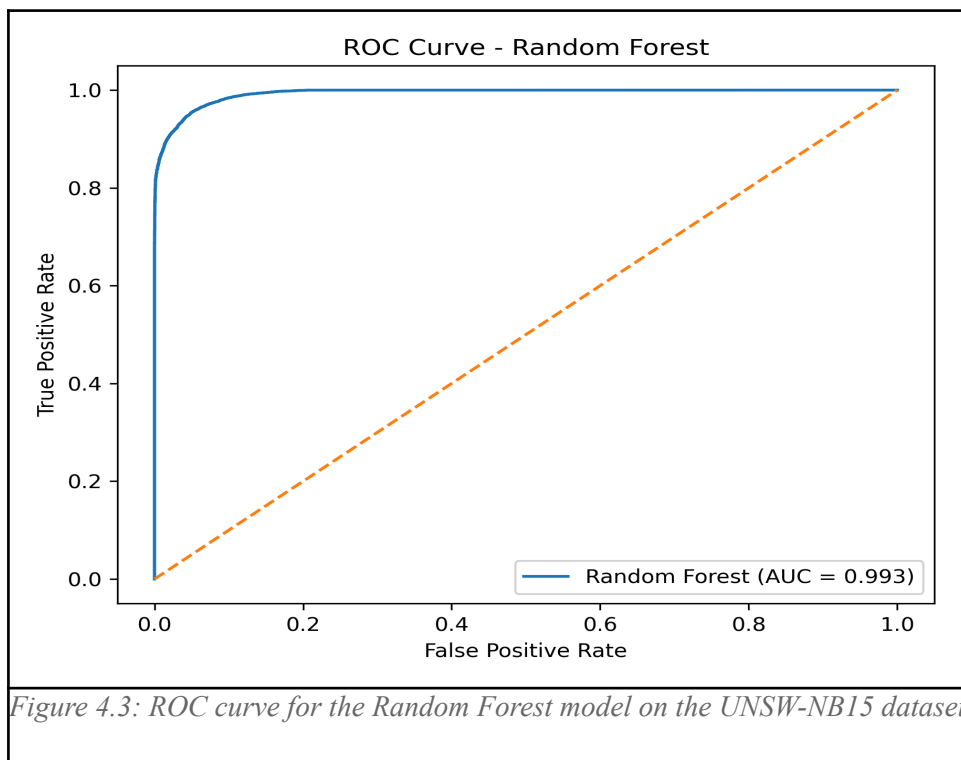
These values show that the model correctly classified a large proportion of both benign and malicious traffic. In particular, the high number of true positives demonstrates that the model was effective at identifying attacks, while the comparatively low number of false negatives suggests that relatively few attacks were missed.

False negatives are especially important in cybersecurity because they represent malicious traffic that escapes detection. Although the Random Forest model did not eliminate false negatives entirely, the value remained low relative to the total number of attack instances. This indicates that the model maintained strong detection capability.

False positives were also considered, as they represent normal traffic incorrectly classified as malicious. In intrusion detection systems, excessive false positives can reduce usability by creating alert fatigue for security analysts. While the model still produced some false alarms, the overall confusion matrix shows that classification performance remained strong and practically useful.

4.5 ROC Curve and AUC for Random Forest

To further evaluate discrimination performance, a Receiver Operating Characteristic (ROC) curve was generated for the Random Forest model. The ROC curve illustrates the relationship between the true positive rate and false positive rate across classification thresholds, while the Area Under the Curve (AUC) provides a single summary value of discrimination performance.



The Random Forest model achieved a **ROC-AUC score of 0.993702**, which indicates excellent separation between benign and malicious traffic. The ROC curve remained very close to the top-left corner of the graph, showing that the model maintained a high true positive rate while keeping the false positive rate relatively low across thresholds.

This result is significant because AUC values close to 1.0 indicate very strong classification ability. In the context of intrusion detection, this suggests that the Random Forest model is highly effective not

only at a single threshold, but across a wide range of possible operating points. This makes the model more robust and more suitable for real-world scenarios where the trade-off between sensitivity and false alarms may need to be adjusted.

4.6 False Positive Rate

The False Positive Rate (FPR) was calculated specifically for the Random Forest model using the confusion matrix values. The model produced a **false positive rate of 8.7727%**.

This metric is important because it reflects how often normal traffic is incorrectly identified as malicious. In practical intrusion detection systems, a lower false positive rate is desirable because it reduces unnecessary alerts and helps security analysts focus on genuinely suspicious activity.

Although the Random Forest model produced some false positives, the value remains acceptable when considered alongside the model's very strong recall, precision, and ROC-AUC. This suggests that the model was able to maintain strong detection performance while keeping false alarms within a manageable range.

4.7 Feature Importance Analysis

One of the key advantages of the Random Forest classifier is that it provides feature importance estimates. These values indicate which input variables contributed most strongly to the model's decisions.

The top ten most important features in the final Random Forest model were:

- `sttl`
- `ct_state_ttl`
- `dload`
- `dttl`
- `rate`
- `ackdat`
- `sload`
- `dmean`
- `tcprrt`
- `sbytes`

Among these, `sttl` and `ct_state_ttl` were the most influential features by a clear margin. This suggests that packet time-to-live related behaviour and state-based traffic characteristics were particularly important for distinguishing malicious traffic from benign traffic in the dataset. Other influential variables such as `dload`, `rate`, `sload`, and `sbytes` indicate that traffic load, communication rate, and byte-level characteristics also played an important role in identifying intrusions.

These results strengthen the interpretability of the model by showing that the classifier relied on meaningful network traffic features rather than arbitrary identifiers. From a cybersecurity perspective, this is valuable because it demonstrates that the model is learning relevant behavioural characteristics associated with attack activity.

4.8 Comparative Performance of Machine Learning Models

To determine whether Random Forest was the most suitable classifier for this task, two additional models were evaluated: Logistic Regression and Support Vector Machine (SVM). These models were selected to provide comparison across different types of machine learning approaches.

The results of the model comparison are shown below:

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Random Forest	0.958026	0.956857	0.982720	0.969616	0.993702
Logistic Regression	0.932276	0.916006	0.991548	0.952281	0.976403
SVM	0.931335	0.910748	0.996946	0.951900	0.975732

Table 4.1: Performance comparison of Random Forest, Logistic Regression, and SVM.

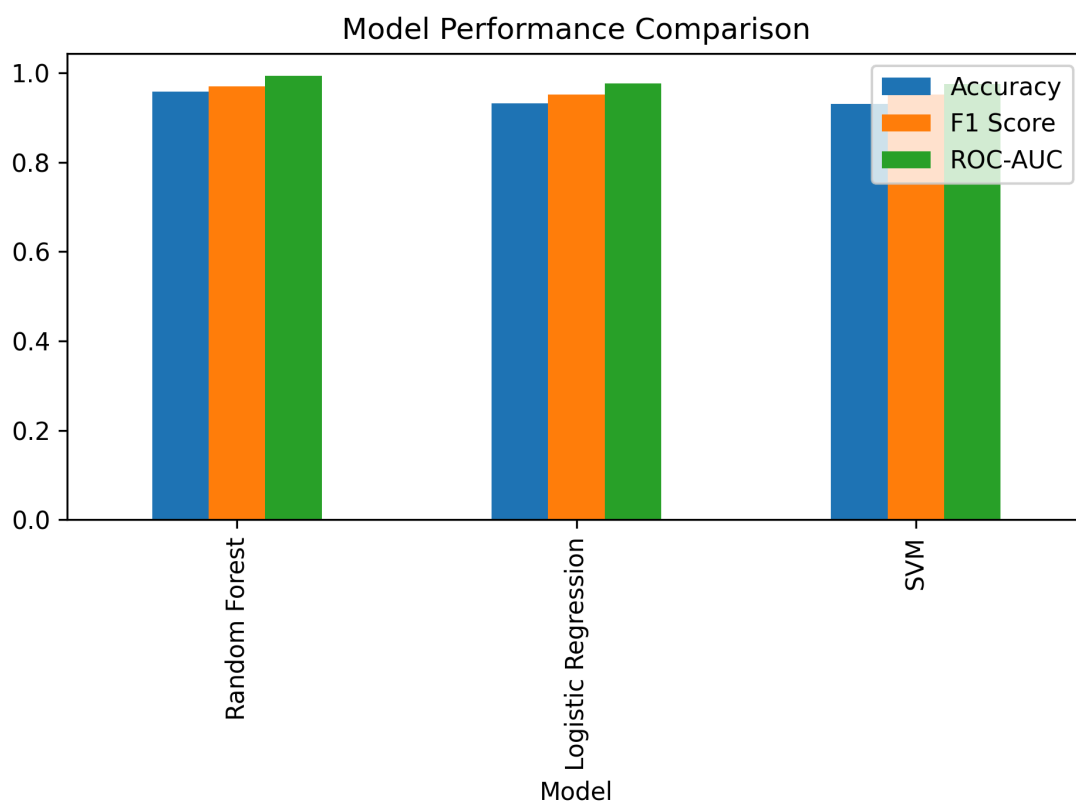


Figure 4.4: Comparison of model performance across Accuracy, F1-Score, and ROC-AUC.

These results show that **Random Forest achieved the strongest overall performance**, with the highest accuracy, precision, F1-score, and ROC-AUC. This indicates that it provided the best overall balance between correct detection and misclassification control.

Logistic Regression and SVM both performed well, particularly in terms of recall. In fact, SVM achieved the **highest recall (0.996946)** among all models, meaning it detected the greatest proportion of attack traffic. However, this came at the cost of lower precision and lower overall accuracy compared with Random Forest. This suggests that SVM was more aggressive in identifying attacks, but also more likely to misclassify benign traffic.

Logistic Regression also produced strong recall, but its overall performance remained below Random Forest. This implies that the intrusion detection problem involves patterns that are not fully captured by simpler linear decision boundaries.

4.9 ROC Curve Comparison Across Models

A combined ROC comparison plot was generated for Random Forest, Logistic Regression, and SVM. This provided a visual comparison of each model's discrimination ability across classification thresholds.

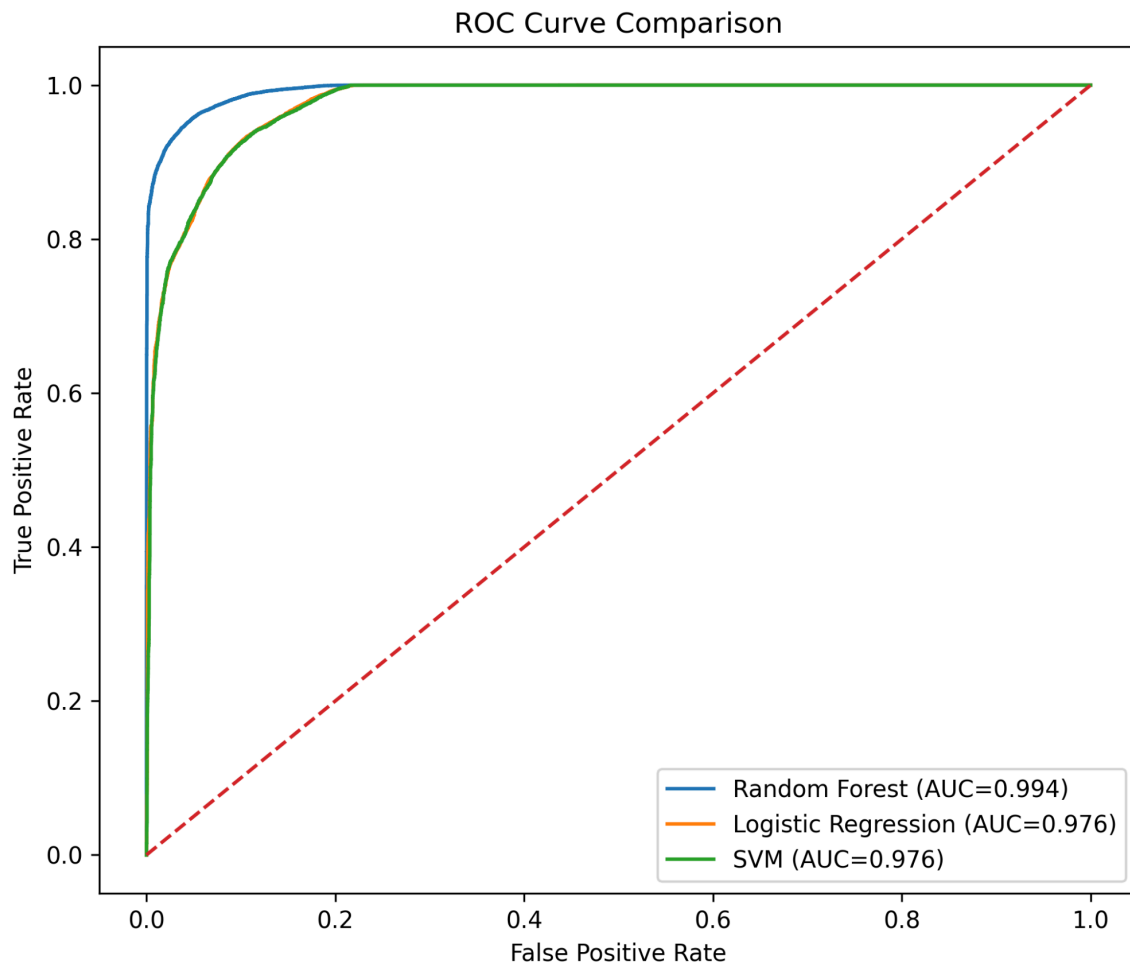


Figure 4.5: ROC curve comparison of Random Forest, Logistic Regression, and SVM.

The ROC comparison showed that the **Random Forest curve remained closest to the top-left corner**, indicating the strongest overall discrimination performance. Its **AUC of 0.993702** was higher than both Logistic Regression (**0.976403**) and SVM (**0.975732**).

Although the Logistic Regression and SVM curves were also strong and substantially above the random baseline, both remained below the Random Forest curve throughout most of the graph. This indicates that while all three models were effective intrusion detectors, Random Forest was consistently better at separating attack traffic from normal traffic across thresholds.

The ROC comparison is particularly useful because it confirms that Random Forest's stronger performance is not limited to one threshold setting, but is maintained across different operating conditions. This strengthens the conclusion that it is the most robust model of the three tested.

4.10 Identification of the Best Model

Based on the combined evidence from the classification metrics, confusion matrix, false positive rate, ROC curve, and model comparison results, **Random Forest was identified as the best-performing model in this study.**

Although SVM achieved the highest recall, Random Forest achieved the best overall balance across all major evaluation measures. It produced the highest accuracy, the highest precision, the highest F1-score, and the highest ROC-AUC. It also provided the added advantage of feature importance analysis, which improved interpretability.

This result suggests that ensemble tree-based learning is particularly effective for the UNSW-NB15 binary intrusion detection task. The model appears to benefit from its ability to capture complex non-linear interactions among network traffic features, something that simpler linear methods may not represent as effectively.

4.11 Summary of Results

The results of the experiments demonstrate that machine learning models can effectively detect malicious network traffic in the UNSW-NB15 dataset. All three evaluated models performed strongly, but their strengths differed across evaluation measures.

The key findings of this chapter can be summarised as follows:

- the Random Forest model achieved strong binary intrusion detection performance with an accuracy of **0.958026** and ROC-AUC of **0.993702**
- the confusion matrix showed high correct classification rates for both normal and attack traffic
- the Random Forest model produced a false positive rate of **8.7727%**
- feature importance analysis showed that the model relied on meaningful traffic features such as `sttl`, `ct_state_ttl`, `dload`, and `dttl`
- Logistic Regression and SVM both produced very high recall, with SVM achieving the highest recall overall
- Random Forest outperformed Logistic Regression and SVM in terms of overall balance, achieving the strongest combined performance across accuracy, precision, F1-score, and ROC-AUC

These findings provide the empirical basis for the next chapter, which discusses why the models performed differently, what the results imply for intrusion detection practice, and what limitations should be considered when interpreting the study.

Chapter 5: Discussion

5.1 Introduction

This chapter discusses the findings presented in Chapter 4 and interprets their significance in relation to the project aim. The purpose of this study was to investigate the use of machine learning for network intrusion detection using the UNSW-NB15 dataset and to compare the performance of different non-LLM models in a binary classification setting. The discussion focuses on why the models performed differently, what the results imply for practical intrusion detection, and what limitations should be considered when interpreting the findings.

5.2 Overall Effectiveness of Machine Learning for Intrusion Detection

The results of this study show that machine learning can be highly effective for detecting malicious network traffic. All three evaluated models — Random Forest, Logistic Regression, and SVM — achieved strong performance across multiple evaluation metrics. This indicates that the UNSW-NB15 dataset contains patterns that can be successfully learned by supervised machine learning algorithms and that non-LLM techniques remain highly relevant in cybersecurity applications.

The strongest evidence for this comes from the consistently high recall and ROC-AUC values across all models. Even the weakest-performing model in overall terms still achieved a ROC-AUC above 0.97, suggesting that the classification task was highly learnable using the selected features. This supports the broader argument of the thesis that machine learning can provide effective support for intrusion detection by identifying behavioural differences between normal and malicious traffic (Kaur, Gabrijelčič, and Klobučar, 2023).

At the same time, the comparison between models shows that not all machine learning techniques are equally suitable. While all three models performed well, the differences between them demonstrate that model choice has a significant effect on the balance between detection performance and false alarms.

5.3 Why Random Forest Performed Best Overall

Random Forest achieved the strongest overall performance in this study, with the highest accuracy, precision, F1-score, and ROC-AUC. This suggests that it was the most effective model for balancing correct attack detection with overall classification reliability.

One reason for this is that Random Forest is well suited to tabular datasets with many interacting features. Network intrusion datasets such as UNSW-NB15 contain complex relationships between packet-level statistics, traffic rates, protocol behaviour, and state information. These relationships are unlikely to be purely linear. Random Forest is able to model such interactions more effectively than linear models because it builds multiple decision trees and combines their outputs. This enables it to capture non-linear decision boundaries and subtle combinations of features associated with attacks.

Another important factor is the robustness of ensemble learning. By averaging the results of multiple trees, Random Forest reduces the risk of overfitting to noise in the training data while still preserving strong predictive power. This is particularly useful in cybersecurity datasets, where patterns may be noisy, heterogeneous, and distributed across many variables.

The high ROC-AUC score achieved by Random Forest also suggests that the model's superiority was not limited to a single decision threshold. Instead, it maintained strong discrimination between malicious and benign traffic across a wide range of thresholds. This makes it not only the best-performing model in terms of static metrics, but also the most robust overall classifier among those evaluated.

5.4 Why SVM Achieved the Highest Recall

Although Random Forest performed best overall, SVM achieved the highest recall. This means that it detected the greatest proportion of malicious instances and missed the fewest attacks. In intrusion detection, this is a valuable property because false negatives represent malicious traffic that escapes detection.

The high recall of SVM suggests that it was more aggressive in labelling traffic as malicious. This may be advantageous in environments where the cost of missing attacks is especially high. However, this stronger sensitivity came at the expense of lower precision and lower overall accuracy compared with Random Forest. In other words, the SVM model was better at capturing nearly all attack instances, but it was less selective and therefore more likely to classify normal traffic incorrectly.

This trade-off is important in real-world cybersecurity. A model with extremely high recall may be desirable in critical environments where missed attacks are unacceptable, such as military, financial, or healthcare systems. However, if that performance comes with too many false alarms, the system may become difficult to use in practice. Analysts may lose confidence in alerts if too many benign events are flagged as malicious.

The SVM results therefore highlight an important point: the “best” model depends partly on operational priorities. If maximum attack capture is the main objective, SVM has an advantage. If a more balanced and operationally practical system is required, Random Forest is more suitable.

5.5 Logistic Regression as a Baseline Model

Logistic Regression performed slightly better than SVM in accuracy and F1-score, but below Random Forest overall. Its results show that even a relatively simple linear model can perform strongly on this dataset. This is useful because it demonstrates that the intrusion detection problem is not so complex that only advanced ensemble models can achieve useful results.

However, the fact that Logistic Regression did not outperform Random Forest suggests that the decision boundary between normal and malicious traffic is not purely linear. While Logistic Regression can serve as a strong baseline and offers the advantage of simplicity, it is less capable of modelling more complex feature interactions. This is consistent with the nature of network traffic

data, where attack behaviour often emerges through combinations of attributes rather than single independent variables.

The inclusion of Logistic Regression in the study was still valuable because it provided a clear benchmark. Without such a baseline, it would be more difficult to demonstrate that the stronger performance of Random Forest is due to its ability to model complex patterns rather than simply the dataset being easy to classify.

5.6 Interpretation of False Positive Rate

The Random Forest model produced a false positive rate of 8.7727%. This means that a proportion of benign traffic was incorrectly flagged as malicious. In an academic setting, this value is still acceptable when considered alongside the model's strong recall and ROC-AUC. However, in real-world intrusion detection, false positive rate is one of the most practically important metrics.

A system with a high false positive rate can generate alert fatigue, where analysts become overwhelmed by the number of warnings and may begin to ignore alerts altogether. This reduces the usefulness of the system, even if its detection capability is strong (**Sommer and Paxson, 2010**). Therefore, while the Random Forest model performed well overall, the false positive rate suggests that further optimisation would still be necessary before operational deployment.

This result also highlights the importance of interpreting intrusion detection models beyond accuracy alone (**Sommer and Paxson, 2010**). A model may achieve high accuracy simply because it correctly classifies the majority of cases, but this does not necessarily mean it is operationally effective. In cybersecurity, the balance between attack detection and false alarms is a central concern. The results of this study reflect that trade-off clearly.

5.7 Meaning of the Feature Importance Results

The feature importance analysis from the Random Forest model provides useful insight into which network characteristics were most relevant for attack detection. The most influential features included `sttl`, `ct_state_ttl`, `dload`, `dtttl`, `rate`, and `ackdat`, along with other traffic- and timing-related variables.

These results suggest that the model relied heavily on packet time-to-live characteristics, connection state behaviour, and traffic intensity measures. This is reasonable from a cybersecurity perspective. Attack traffic often differs from benign traffic in terms of packet flow behaviour, connection states, and the rate at which data is transmitted. The fact that these types of features emerged as important indicates that the model was capturing meaningful behavioural patterns rather than relying on arbitrary or irrelevant variables.

This improves the interpretability of the study. One common criticism of machine learning in cybersecurity is that high-performing models may behave like black boxes. The feature importance analysis partly addresses this concern by showing that the model's decisions were informed by technically plausible traffic characteristics. While this does not fully explain individual predictions, it does strengthen confidence in the model's overall behaviour.

5.8 Impact of Removing the ‘id’ Feature

An important methodological issue arose during the project when it was discovered that the `id` column had initially been included in the feature set. Since `id` is only a record identifier and not a meaningful traffic feature, it should not be used for prediction. After removing it and rerunning the experiments, the results became slightly weaker but more realistic.

This change was significant because it improved the validity of the study. When `id` was included, it appeared as the most important feature, which suggested that the model was exploiting a non-meaningful pattern in the data. Removing it reduced the apparent performance of the models, particularly Random Forest, but it made the evaluation much more methodologically sound.

This correction ultimately strengthens the dissertation rather than weakening it. It shows critical evaluation of the modelling process and demonstrates awareness of data leakage risks. In a research context, this is important because robust methodology is more valuable than artificially inflated performance. The final results are therefore more trustworthy and easier to defend.

5.9 Relation to the Project Aim

The aim of the project was to investigate how non-LLM machine learning techniques can be used for intrusion detection and to evaluate their effectiveness using benchmark cybersecurity data. The results support this aim strongly.

First, the study demonstrated that multiple machine learning models can successfully classify malicious and benign traffic using the UNSW-NB15 dataset. Second, it showed that different models exhibit different strengths: Random Forest offered the best overall balance, while SVM provided the highest recall. Third, the use of feature importance analysis showed that the selected defensive model could provide not only strong performance but also some interpretability.

Taken together, these findings confirm that machine learning is a useful and effective approach for intrusion detection in controlled experimental settings. They also support the idea that careful model selection is essential, as performance depends not only on whether machine learning is used, but on which algorithm is chosen and what operational trade-offs are prioritised.

5.10 Practical Implications for Cybersecurity

From a practical perspective, the results suggest that Random Forest would be a strong candidate for deployment in a defensive intrusion detection context, especially in environments where balanced performance is required. Its high precision, high recall, and high ROC-AUC make it suitable for identifying attacks while maintaining a reasonable degree of selectivity.

However, the results also show that no model is perfect. Even the strongest model still produced false positives and false negatives. This means that in practice, such a model should be viewed as a support tool rather than a complete replacement for human analysts or other security controls. Machine learning-based detection is best used as part of a broader security architecture that includes logging,

rule-based analysis, expert review, and incident response processes (**Kaur, Gabrijelčič, and Klobučar, 2023**).

The strong recall achieved by SVM also suggests that alternative models may be preferable in high-sensitivity environments where missing attacks is considered less acceptable than generating false alarms. This reinforces the idea that intrusion detection models must be evaluated not only by overall performance but also by their suitability for particular operational contexts.

5.11 Limitations of the Study

Although the findings are strong, several limitations should be acknowledged.

First, the study was conducted using a benchmark dataset rather than live network traffic. While UNSW-NB15 is widely used and more modern than older intrusion detection datasets, it still represents an offline experimental environment (**Moustafa and Slay, 2015**). Real-world networks are more dynamic, noisy, and variable than benchmark datasets, so the observed performance may not translate directly to operational deployment (**Sommer and Paxson, 2010**).

Second, the study focused on binary classification rather than multi-class attack categorisation. This means the models were only required to distinguish between benign and malicious traffic, not between different categories of attacks. A multi-class setting would provide more detailed insight into how well machine learning can differentiate between specific threat types.

Third, only three classical machine learning models were evaluated. While these models are appropriate for the scope of the project, the study does not explore deep learning, hybrid models, or real-time streaming approaches. As a result, the findings are limited to non-LLM classical machine learning techniques.

Fourth, feature encoding was performed using label encoding for categorical variables. While this is efficient and practical, it may not capture all relationships between categorical states as effectively as alternative encoding strategies. This could influence model behaviour and may partly limit performance.

Finally, the project evaluated performance primarily using a single train-test split. Although this is standard and acceptable for the project scope, additional validation approaches such as k-fold cross-validation could provide a more comprehensive assessment of model stability.

5.12 Future Work

Several extensions could improve or expand the study in future work.

A first improvement would be to evaluate the models on additional intrusion detection datasets, such as CICIDS2017, in order to test generalisability across data sources. This would help determine whether the findings are specific to UNSW-NB15 or representative of intrusion detection more broadly (**Moustafa and Slay, 2015; Sharafaldin, Lashkari and Ghorbani, 2018**).

A second extension would be to investigate multi-class classification using the `attack_cat` field. This would allow the study to examine whether different machine learning models are better suited to identifying specific attack types rather than simply distinguishing attack from normal traffic.

A third direction would be to compare the classical models used in this project with more advanced approaches such as gradient boosting, neural networks, or hybrid intrusion detection architectures. This could provide a broader picture of where Random Forest stands relative to more recent methods.

A fourth area of future work would be deployment-oriented testing, such as real-time intrusion detection, streaming traffic classification, or integration with security monitoring tools. This would bring the study closer to operational cybersecurity practice.

5.13 Summary of Discussion

This chapter has shown that the results support the use of machine learning for intrusion detection, while also highlighting important differences between models. Random Forest performed best overall because it provided the strongest balance across accuracy, precision, F1-score, and ROC-AUC. SVM achieved the highest recall, making it particularly effective at capturing malicious traffic, though at the cost of lower precision and overall balance. Logistic Regression provided a strong and useful baseline, but it was less capable of handling the full complexity of the dataset.

The discussion also showed that false positive rate remains a practical concern, that feature importance enhances interpretability, and that removing the `id` feature improved the methodological soundness of the study. Although the findings are promising, the study remains limited by its use of an offline benchmark dataset and a restricted set of classical machine learning models.

Overall, the results suggest that Random Forest is the most suitable model among those tested for binary intrusion detection on UNSW-NB15, while also demonstrating that intrusion detection performance must always be interpreted in relation to operational trade-offs and research limitations.

Chapter 6: Conclusion

6.1 Conclusion

The aim of this project was to investigate the use of non-LLM machine learning techniques for network intrusion detection using the UNSW-NB15 dataset. In particular, the study sought to evaluate how effectively different machine learning models could distinguish between normal and malicious network traffic in a binary classification setting, and to compare their performance using standard intrusion detection metrics.

To achieve this aim, the project followed a structured experimental workflow consisting of data exploration, preprocessing, model training, and comparative evaluation. The UNSW-NB15 dataset was selected because it provides a modern and widely used benchmark for intrusion detection

research. After preprocessing the dataset and removing non-informative attributes, three machine learning models were implemented and evaluated: Random Forest, Logistic Regression, and Support Vector Machine.

The results showed that all three models were capable of detecting malicious traffic with strong overall performance. However, important differences were observed between them. Random Forest produced the strongest overall results, achieving the best balance across accuracy, precision, F1-score, and ROC-AUC. Logistic Regression provided a useful baseline and demonstrated that even a relatively simple linear model can perform effectively on intrusion detection data. SVM achieved the highest recall, indicating particularly strong attack detection sensitivity, although this came with lower overall balance than Random Forest.

These findings suggest that machine learning is a highly effective approach for binary intrusion detection in controlled dataset-based environments. More specifically, the results indicate that ensemble methods such as Random Forest are especially well suited to this task because they can capture non-linear relationships between network traffic features while also providing interpretable feature importance information.

The project also highlighted the importance of careful preprocessing and methodological validation. The removal of the `id` field from the feature set improved the validity of the final pipeline and ensured that the reported results were based on meaningful traffic characteristics rather than record identifiers. This strengthened the reliability of the study and demonstrated the importance of critically reviewing feature selection decisions during machine learning experimentation.

Overall, the project successfully met its aim and objectives. It demonstrated that non-LLM machine learning models can be used effectively for intrusion detection, that model selection has a significant effect on performance, and that Random Forest was the most suitable of the evaluated models for the UNSW-NB15 binary classification task.

6.2 Contributions of the Project

The main contributions of this project can be summarised as follows:

- it developed a complete machine learning pipeline for binary intrusion detection using the UNSW-NB15 dataset
- it evaluated and compared three non-LLM machine learning models in a consistent experimental setting
- it identified Random Forest as the strongest overall model for the dataset and task used in this study
- it provided interpretable feature importance analysis to show which network traffic characteristics contributed most strongly to detection
- it demonstrated the importance of methodological care in preprocessing and feature selection

These contributions support the wider argument that classical machine learning remains highly relevant for cybersecurity defence, particularly for structured network traffic analysis.

6.3 Limitations of the Project

Although the project achieved its objectives, several limitations should be acknowledged. First, the experiments were conducted using an offline benchmark dataset rather than live network traffic. This means that the results reflect controlled experimental conditions and may not fully represent the complexity of operational network environments.

Second, the task was limited to binary classification rather than multi-class attack categorisation. While this was appropriate for the project scope, it does not show how well the models would perform when distinguishing between different types of attacks.

Third, the study focused only on three classical machine learning models. Although these were sufficient for the project aim, more advanced methods such as gradient boosting, deep learning, or hybrid intrusion detection systems were not explored.

Finally, the evaluation was based primarily on a single train-test split, which, while valid for the scope of the project, provides a narrower assessment than repeated validation or cross-validation approaches.

6.4 Future Work

Several directions for future work emerge from this study.

One possible extension would be to evaluate the same models on additional intrusion detection datasets, such as CICIDS2017, in order to assess whether the findings generalise beyond UNSW-NB15. Another useful direction would be to expand the study from binary classification to multi-class attack classification using the `attack_cat` field. This would provide a more detailed understanding of model performance across different threat types.

Further work could also compare the models used in this project with more advanced algorithms, such as gradient boosting methods or neural network approaches. In addition, future research could investigate real-time or streaming intrusion detection in order to move closer to practical deployment scenarios.

Finally, further attention could be given to threshold optimisation and false positive reduction, since operational intrusion detection systems must balance strong attack detection with manageable alert volumes.

6.5 Final Statement

In conclusion, this project demonstrates that non-LLM machine learning techniques can provide highly effective support for intrusion detection when applied to structured network traffic data. Among the models tested, Random Forest emerged as the most effective overall solution, offering the best combination of predictive performance, robustness, and interpretability. While further work is needed to assess deployment in real-world environments, the study provides strong evidence that machine learning remains a valuable defensive tool in modern cybersecurity.

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